

Week 8 Tutorial

Enterprise Valuation

Zirui Song
April 2025



Agenda

- 9-2
- 9-3
- 9-5

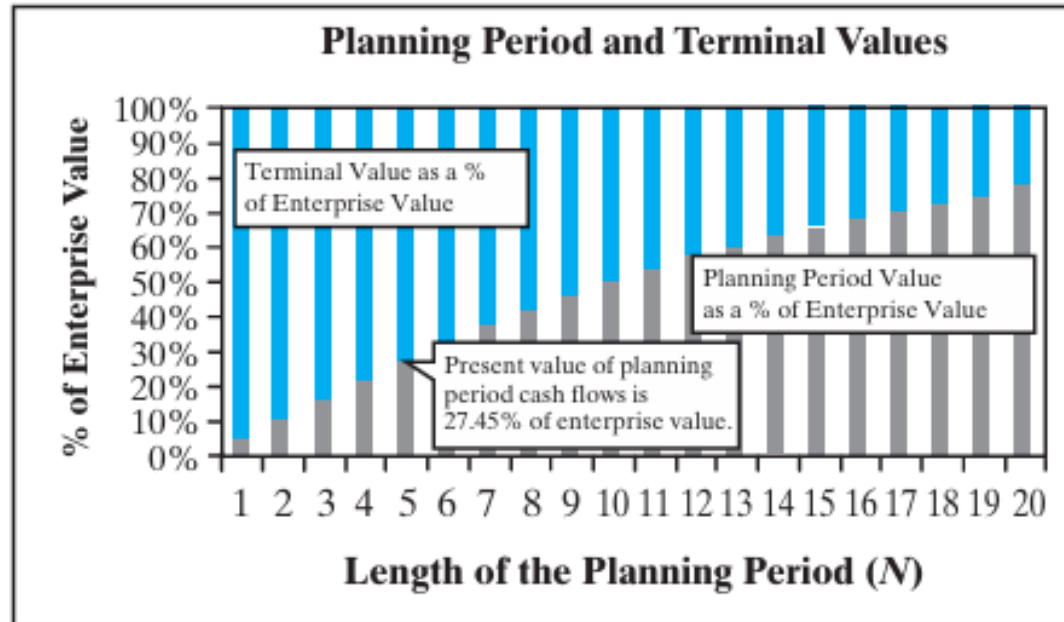
Review 1

$$\begin{array}{l} \text{Enterprise} \\ \text{Value} \end{array} = \begin{array}{l} \text{Present Value} \\ \text{of the Planning} \\ \text{Period (PP) Cash Flows} \end{array} + \begin{array}{l} \text{Present Value} \\ \text{of the Terminal} \\ \text{Value} \end{array} \quad (9.1)$$

- The traditional WACC method
- What proportion of the enterprise value do you think comes from the terminal value here?
- Answer: **Over 50%!**

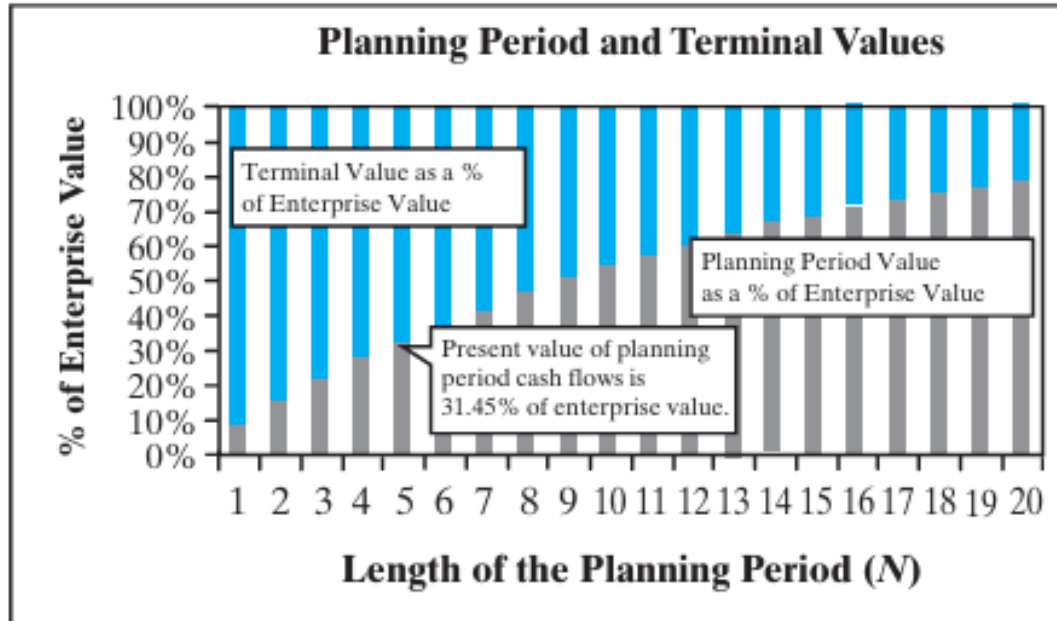
Review 1

- Consider the situation where a firm's FCFs are expected to grow at a rate of 12% per year for a period of 5 years, followed by 2% rate of growth thereafter. And the cost of capital for this firm is 10%.



Review 1

- If we reduce the growth rate from 12% to 2%, indicating a stable (low-growth) firm, the terminal value is still the dominant component of the enterprise value estimates for the typical three- to ten- planning period.



Review 1

$$\text{Enterprise Value} = \text{Present Value of the Planning Period (PP) Cash Flows} + \text{Present Value of the Terminal Value} \quad (9.1)$$

$$\text{Firm Value}_0 = \sum_{t=1}^N \frac{E(\text{Firm FCF}_t)}{(1 + \text{WACC})^t} \quad (4.2)$$

$$\text{WACC} = k_d(1 - T)w_d + k_p w_p + k_e w_e \quad (4.1)$$

$$\text{Terminal Value}_{2020} = \frac{\text{FCF}_{2020} \left(1 + \frac{\text{Terminal Growth Rate}(g)}{\text{Cost of Capital} (k_{\text{WACC}} - \text{Terminal Growth Rate}(g))} \right)}{\quad} \quad (9.2)$$

g?

$$\text{Terminal Value}_{2020} = \text{FCF}_{2020} \left(\frac{1 + g}{k_{\text{WACC}} - g} \right) = \text{FCF}_{2020} \times \left(\text{Gordon Growth Model Multiple} \right)$$

$$\text{Terminal Value}_{2020} = \text{EBITDA}_{2020} \times \text{EBITDA Multiple} \quad (9.3)$$

Review 1



WORLD BANK GROUP

Data

GDP growth (annual %)

Australia

Search data e.g. GDP, population, Indonesia

GDP growth (annual %) - Australia

World Bank national accounts data, and OECD National Accounts data files.

License : CC BY-4.0

Line

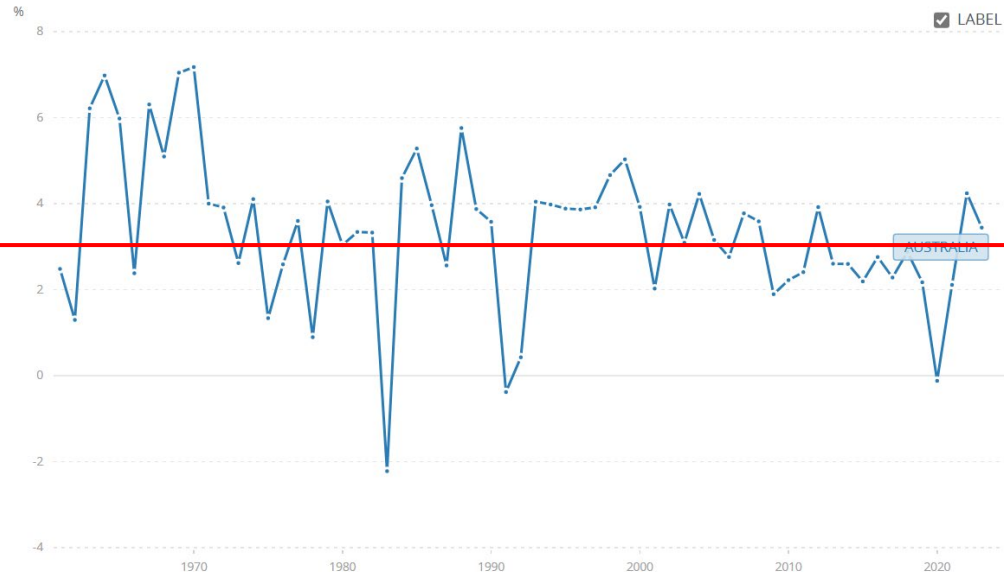
Bar

Map

Also Show

Share

Details



Review 1

- $TV_{2020} = \frac{FCF_{2020} \times (1+g)}{(1+WACC)^1} + \frac{FCF_{2020} \times (1+g)^2}{(1+WACC)^2} + \dots$ $n = \infty$
- *Let the common ratio (r) be $\frac{1+g}{1+WACC}$, and the first term (a) is $FCF_{2020} \times r$*
- Now, the Gordon Growth Model is based on the **infinite geometric series**.
- $TV_{2020} = a + ar + ar^2 + \dots$ (1)
- Multiply both sides by r: $r \times TV_{2020} = ar + ar^2 + \dots$ (2)
- Subtract equation (2) from equation (1): $TV_{2020} - r \times TV_{2020} = a$
- Solve for TV_{2020} : $TV_{2020} = \frac{a}{1-r} = \frac{FCF_{2020} \times r}{1 - \frac{1+g}{1+WACC}} = \frac{FCF_{2020} \times \frac{1+g}{1+WACC}}{1 - \frac{1+g}{1+WACC}} = \frac{FCF_{2020}(1+g)}{WACC-g}$
- No-growth scenario ($g = 0$): $TV_{2020} = \frac{FCF_{2020}}{WACC}$

Review 1: GRC acquisition

Table 9-1 Pre- and Postacquisition Balance Sheets for GRC

	Preacquisition 2014	Postacquisition 2014
Current assets	\$ 41,865,867	\$ 41,865,867
Gross property, plant, and equipment	88,164,876	88,164,876
Less: accumulated depreciation	(41,024,785)	(41,024,785)
Net property, plant, and equipment	\$ 47,140,092	\$ 47,140,092
Goodwill	—	20,819,867
Total	\$ 89,005,959	\$ 109,825,826
Current liabilities	\$ 9,825,826	\$ 9,825,826
Long-term debt	36,839,923	40,000,000
Total liabilities	\$ 46,665,749	\$ 49,825,826
Common stock (par)	290,353	400,000
Paid-in capital	20,712,517	59,600,000
Retained earnings	21,337,340	—
Common equity	\$ 42,340,210	\$ 60,000,000
Total	\$ 89,005,959	\$ 109,825,826

$$=60,000,000-(89,005,959-49,825,826)$$

Immersion can acquire GRC for a cash payment of \$100 million, which is equal to a multiple of six times GRC's 2014 EBITDA. Immersion will also be assuming GRC's accounts payable and accrued expenses, which total \$9,825,826, so the book value of the GRC acquisition would be \$109,825,826 (see Table 9-1). To finance the GRC acquisition, Immersion will borrow \$40 million in nonrecourse debt and will need to raise \$60 million in equity.¹ Based on conversations with its investment

Acquisition financing—

Assumed payables and accrued expenses	\$ 9,825,826
Long-term debt (40% of enterprise value)	40,000,000
Sale of common stock	60,000,000
Total	\$ 109,825,826
Shares issued (\$1.00 par value)	400,000
Net proceeds	\$ 60,000,000
Price per share	\$ 150.00

Review 1: GRC acquisition

Table 9-2 Estimating GRC's Enterprise Value Using DCF Analysis (Status Quo Strategy)

Panel (a) Step 1: Estimate the Amount and Timing of the Planning Period Future Cash Flows

Planning Period Pro Forma Financial Statements

Assumption: a constant growth rate of 4%

	Preacquisition	Postacquisition	Pro Forma Income Statements					
	2014	2014	2015	2016	2017	2018	2019	2020
Revenues	\$ 80,000,000	\$ 80,000,000	\$ 83,200,000	\$ 86,528,000	\$ 89,989,120	\$ 93,588,685	\$ 97,332,232	\$ 101,225,521
Cost of goods sold	(45,733,270)	(45,733,270)	(50,932,000)	(52,629,280)	(54,394,451)	(56,230,229)	(58,139,438)	(60,125,016)
Gross profit	\$ 34,266,730	\$ 34,266,730	\$ 32,268,000	\$ 33,898,720	\$ 35,594,669	\$ 37,358,456	\$ 39,192,794	\$ 41,100,506
General and administrative expense	(17,600,000)	(17,600,000)	(17,984,000)	(18,383,360)	(18,798,694)	(19,230,642)	(19,679,868)	(20,147,063)
Depreciation expense	(6,500,000)	(6,500,000)	(7,856,682)	(7,856,682)	(7,856,682)	(7,856,682)	(7,856,682)	(7,856,682)
Net operating income	\$ 10,166,730	\$ 10,166,730	\$ 6,427,318	\$ 7,658,678	\$ 8,939,292	\$ 10,271,131	\$ 11,656,244	\$ 13,096,761
Interest expense	(2,523,020)	(2,523,020)	(2,600,000)	(2,549,847)	(2,453,680)	(2,307,941)	(2,108,865)	(1,852,465)
Earnings before taxes	\$ 7,643,710	\$ 7,643,710	\$ 3,827,318	\$ 5,108,831	\$ 6,485,612	\$ 7,963,190	\$ 9,547,379	\$ 11,244,296
Taxes	(1,910,928)	(1,910,928)	(956,830)	(1,277,208)	(1,621,403)	(1,990,798)	(2,386,845)	(2,811,074)
Net income	\$ 5,732,783	\$ 5,732,783	\$ 2,870,489	\$ 3,831,623	\$ 4,864,209	\$ 5,972,393	\$ 7,160,534	\$ 8,433,222

Review 1: GRC acquisition

Assumption: increases in proportion to firm revenues

	Preacquisition	Postacquisition	Pro Forma Balance Sheets					
	2014	2014	2015	2016	2017	2018	2019	2020
Current assets	\$ 41,865,867	\$ 41,865,867	43,540,502	45,282,122	47,093,407	48,977,143	50,936,229	52,973,678
Gross property, plant, and equipment	88,164,876	88,164,876	96,021,558	103,878,240	111,734,922	119,591,604	127,448,286	135,304,968
Less: accumulated depreciation	(41,024,785)	(41,024,785)	(48,881,467)	(56,738,149)	(64,594,831)	(72,451,513)	(80,308,194)	(88,164,876)
Net property, plant, and equipment	\$ 47,140,092	\$ 47,140,092	\$ 47,140,092	\$ 47,140,092	\$ 47,140,092	\$ 47,140,092	\$ 47,140,092	\$ 47,140,092
Goodwill	—	20,819,867	20,819,867	20,819,867	20,819,867	20,819,867	20,819,867	20,819,867
Total	\$ 89,005,959	\$ 109,825,826	\$ 111,500,461	\$ 113,242,081	\$ 115,053,366	\$ 116,937,102	\$ 118,896,188	\$ 120,933,637
Current liabilities	\$ 9,825,826	\$ 9,825,826	\$ 9,975,651	\$ 10,131,469	\$ 10,293,520	\$ 10,462,053	\$ 10,637,327	\$ 10,819,613
Long-term debt	36,839,923	40,000,000	39,228,419	37,748,922	35,506,789	32,444,078	28,499,462	23,608,049
Total liabilities	\$ 46,665,749	\$ 49,825,826	\$ 49,204,070	\$ 47,880,392	\$ 45,800,309	\$ 42,906,131	\$ 39,136,790	\$ 34,427,661
Common stock (par)	290,353	400,000	400,000	400,000	400,000	400,000	400,000	400,000
Paid-in capital	20,712,517	59,600,000	59,600,000	59,600,000	59,600,000	59,600,000	59,600,000	59,600,000
Retained earnings	21,337,340	—	2,296,391	5,361,689	9,253,057	14,030,971	19,759,398	26,505,976
Common equity	\$ 42,340,210	\$ 60,000,000	\$ 62,296,391	\$ 65,361,689	\$ 69,253,057	\$ 74,030,971	\$ 79,759,398	\$ 86,505,976
Total	\$ 89,005,959	\$ 109,825,826	\$ 111,500,461	\$ 113,242,081	\$ 115,053,366	\$ 116,937,102	\$ 118,896,188	\$ 120,933,637

Review 1: GRC acquisition

Planning Period Cash Flow Estimates

Pro forma income statements

Projected Firm FCFs

	2015	2016	2017	2018	2019	2020
Net operating income	\$ 6,427,318	7,658,678	\$ 8,939,292	\$ 10,271,131	\$ 11,656,244	\$ 13,096,761
Less: taxes	(1,606,830)	(1,914,670)	(2,234,823)	(2,567,783)	(2,914,061)	(3,274,190)
Net operating profit after taxes (NOPAT)	\$ 4,820,489	\$ 5,744,009	\$ 6,704,469	\$ 7,703,349	\$ 8,742,183	\$ 9,822,571
Plus: depreciation	7,856,682	7,856,682	7,856,682	7,856,682	7,856,682	7,856,682
Less: capital expenditures (CAPEX)	(7,856,682)	(7,856,682)	(7,856,682)	(7,856,682)	(7,856,682)	(7,856,682)
Less: changes in net working capital	(1,281,602)	(1,332,866)	(1,386,180)	(1,441,628)	(1,499,293)	(1,559,264)
Equals firm FCF	\$ 3,538,887	\$ 4,411,143	\$ 5,318,289	\$ 6,261,721	\$ 7,242,890	\$ 8,263,306

Pro forma balance sheets

Panel (b) Step 1 (continued): Terminal-Value Cash Flow Estimate

Method 1—DCF Using the Gordon Growth Model

Terminal-Value Multiples from the Gordon Model

Terminal-Value Estimates (for FCFs Received in 2021 and Beyond)

Discount Rates	Growth Rates (g)				Discount Rates	Growth Rates (g)			
	0%	1%	2%	3%		0%	1%	2%	3%
8.3%	12.05	13.84	16.19	19.43	8.3%	\$ 99,557,908	\$ 114,327,938	\$ 133,786,865	\$ 160,588,784
8.8%	11.36	12.95	15.00	17.76	8.8%	\$ 93,901,209	\$ 106,999,224	\$ 123,949,596	\$ 146,744,924
9.3%	10.75	12.17	13.97	16.35	9.3%	\$ 88,852,757	\$ 100,553,487	\$ 115,459,897	\$ 135,098,501
9.8%	10.20	11.48	13.08	15.15	9.8%	\$ 84,319,453	\$ 94,840,221	\$ 108,058,622	\$ 125,164,788

Review 1: GRC acquisition

Method 2—Multiples Using Enterprise-Value to EBITDA Ratio

Terminal-Value Estimates

<u>Enterprise Value</u> EBITDA	<u>Terminal</u> Value
5.00	\$104,767,215
5.50	115,243,936
6.00	125,720,658
6.50	136,197,379
7.00	146,674,101

Panel (c) Step 2: Estimate a Risk-Appropriate Discount Rate

- *Cost of debt*—The estimated borrowing rate is 6.5% with a marginal tax of 25%, resulting in an after-tax cost of debt of 4.875%.
- *Cost of equity*—An average industry unlevered equity beta of .89 implies a levered equity beta for GRC of 1.28, assuming a target debt ratio of 40% and a debt beta of .30. Using the capital asset pricing model (CAPM) with a ten-year Treasury bond yield of 5.02% and a market risk premium of 5% produces an estimate of the levered cost of equity of 11.42%.
- *Weighted average cost of capital (WACC)*—Using the target debt-to-value ratio of 40%, the WACC is approximately 8.8%.

Panel (d) Step 3: Calculate the Present Value of Future Cash Flows

Discount Rate	Planning Period FCF	Present Values of Expected Future Cash Flows			
		Terminal Value		Enterprise Value	
		Method 1	Method 2	Method 1	Method 2
7.8%	\$26,201,884	\$78,986,073	\$80,112,266	\$105,187,958	\$106,314,150
8.8%	\$25,309,857	\$74,729,086	\$75,794,582	\$100,038,943	\$101,104,439
9.8%	\$24,461,738	\$70,737,378	\$71,745,960	\$ 95,199,116	\$ 96,207,698

NPV≈0

Problem 1

9-2 ENTERPRISE VALUATION In summer 2015, Smidgeon Industries was evaluating whether to purchase one of its suppliers. The supplier, Carswell Manufacturing, provides Smidgeon with the raw steel Smidgeon uses to fabricate utility trailers. One of the first things that Smidgeon's managers did was to forecast the cash flows of Carswell for the next five years:

Planning period (PP)=5

Year	Cash Flows
1	\$1,200,000
2	1,260,000
3	1,323,000
4	1,389,150
5	1,458,608

Terminal value (TV)= 5 1,458,608 **X 6**

Next, Smidgeon's management team looked at a group of similar firms and estimated Carswell's cost of capital to be 15%. Finally, the team estimated that Carswell would be worth approximately six times its year 5 cash flow in five years.

- What is your estimate of the enterprise value of Carswell?
- What is the value of the equity of Carswell if the acquisition goes through, and Smidgeon borrows \$2.4 million and finances the remainder using equity?

Problem 1

Answer

$$\text{Terminal Value}_{2020} = \text{FCF}_{2020} \left(\frac{1+g}{k_{\text{WACC}} - g} \right) = \text{FCF}_{2020} \times \left(\text{Gordon Growth Model Multiple} \right)$$

$$\text{Enterprise Value} = \text{Present Value of the Planning Period (PP) Cash Flows} + \text{Present Value of the Terminal Value}$$

1. Step 1: TV cash flows estimate

- $TV_5 = 1,458,608 \times 6 = 7,293,040$

Step 2: Estimate a risk-appropriate discount rate (WACC=15%)

Step 3: Calculate the PV of future cash flows (*NPV function from Excel*)

- $$EV = PP_0 + TV_0 = \frac{1,200,000}{(1+0.15)^1} + \frac{1,260,000}{(1+0.15)^2} + \frac{1,323,000}{(1+0.15)^3} + \frac{1,389,150}{(1+0.15)^4} + \frac{1,458,608}{(1+0.15)^5} + \frac{7,293,040}{(1+0.15)^5} = 8,736,664.28$$

Problem 1

2. *Enterprise value = Debt value + Equity value*

- *Equity value = 8,736,664.28 - 2,400,000 = 6,336,664.28*

Review 2: APV

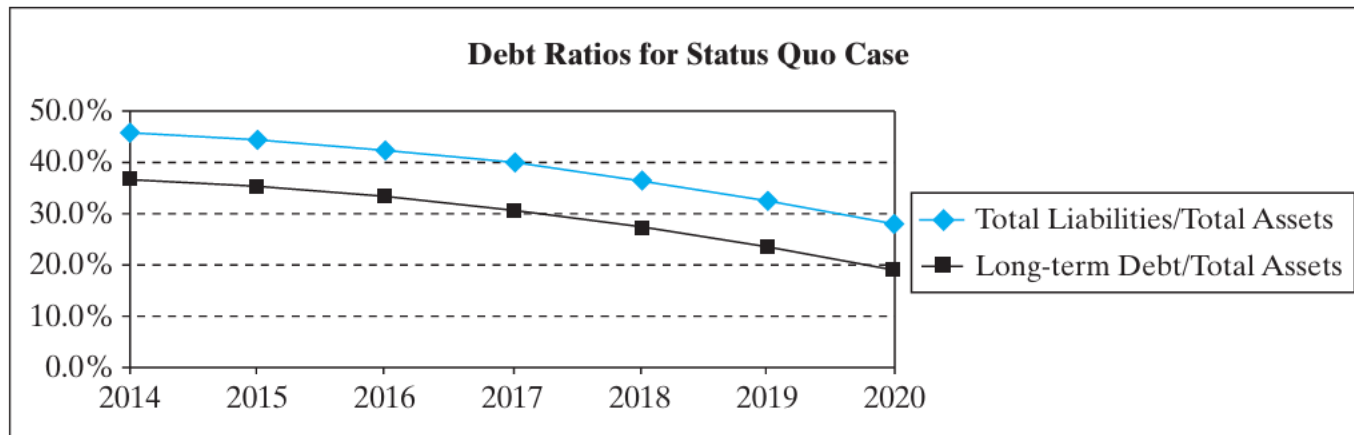
	Adjusted Present Value (APV) Method	Traditional WACC Method
Object of the analysis	<i>Enterprise value</i> as the sum of the values of: ■ The unlevered equity cash flows, and ■ Financing side effects	<i>Enterprise value</i> equal to the present value of the firm's free cash flows, discounted using the after-tax WACC.
Cash flow calculation	■ Unlevered free cash flows (equals firm FCFs), plus ■ Interest tax savings	Firm FCFs.
Discount rate(s)	■ Unlevered cash flows—cost of equity for unlevered firm, and ■ Interest tax savings—the yield to maturity on the firm's debt	After-tax WACC.
How capital structure effects are dealt with—discount rates, cash flows, or both	<i>Cash flows</i> —Capital structure affects the present value of the interest tax savings only. The value of the unlevered firm is not affected by the firm's use of debt financing.	<i>Discount rate</i> —The debt-equity mix affects the firm's WACC. However, the mix of debt and equity is assumed to be constant throughout the life of the investment.

Review 2: APV

Technical Issues – Equations		
Valuation models	<i>Enterprise Value</i> $\begin{aligned} \text{Enterprise Value} &= \text{Value of the Unlevered Firm} + \text{Value of Interest Tax Savings} \\ &= \sum_{t=1}^{\infty} \frac{\text{FCF}_t^{\text{Unlevered}}}{(1 + k_{\text{Unlevered}})^t} \\ &\quad + \sum_{t=1}^{\infty} \frac{\text{Interest} \times \text{Tax Rate}}{(1 + k_d)^t} \end{aligned}$ where $\text{FCF}_t^{\text{Unlevered}}$ = FCFs for an unlevered firm (equals FCF_t). $k_{\text{Unlevered}}$ = cost of equity for an unlevered firm	<i>Enterprise Value</i> $\text{Enterprise Value} = \sum_{t=1}^{\infty} \frac{\text{FCF}_t}{(1 + k_{\text{WACC}})^t}$ where FCF_t = firm FCF_t for year_t k_{WACC} = the firm's WACC
User's Guide		
Choosing the right model	Although not as popular as the traditional WACC method, the APV method is gaining support and is particularly attractive when valuing highly leveraged transactions, <u>such as leveraged buyouts, where capital structure changes through time.</u> Leveraged buyout (LBO)	Most widely used method for valuing individual investment projects (capital budgeting) and entire firms. <u>Because the discount rate (WACC) assumes constant weights for debt and equity,</u> this method is not well suited for situations where dramatic changes in capital structure are anticipated (e.g., leverage buyouts).

Review 2: APV

Figure 9-1 Capital Structure Ratios for the GRC Acquisition, 2014 to 2020



<i>Status Quo Case</i>							
	2014	2015	2016	2017	2018	2019	2020
Total Liabilities/Total Assets	45.4%	44.1%	42.3%	39.8%	36.7%	32.9%	28.5%
Long-term Debt/Total Assets	36.4%	35.2%	33.3%	30.9%	27.7%	24.0%	19.5%

Review 2: APV

Table 9-4 GRC's Operating and Financial Cash Flows for the Planning Period

Panel (a) Unlevered Free Cash Flows (same as FCF)

	2015	2016	2017	2018	2019	2020
Status quo strategy	\$3,538,887	\$4,411,143	\$5,318,289	\$6,261,721	\$7,242,890	\$ 8,263,306
Growth strategy	\$ (104,715)	\$1,483,309	\$3,208,375	\$7,331,446	\$9,364,362	\$11,569,912

Panel (b) Interest Tax Savings

	2015	2016	2017	2018	2019	2020
Status quo strategy	\$ 650,000	\$ 637,462	\$ 613,420	\$ 576,985	\$ 527,216	\$ 463,116
Growth strategy	\$ 650,000	\$ 694,742	\$ 721,884	\$ 729,061	\$ 684,432	\$ 613,272

Problem 2

9-3 ENTERPRISE VALUATION USING THE APV MODEL You have been recently hired as a financial analyst. When you arrive at work on Monday morning, the CFO of the firm sends you a message asking you to step into her office as soon as you have a free minute. Having been on the job just two weeks, you quickly decide you have a free minute and go to the executive floor, where the CFO has her office. You discover that the CFO wants you to educate her on the meaning and use of the APV valuation method. Although she does not elaborate on the reason for her request, you are aware of the office scuttlebutt that the board of directors is considering the prospect of taking the firm private and is in discussions with an investment banker about the particulars of the deal.

After leaving the CFO's office, you decide that the best way to explain the use of the APV methodology is to construct a simple example. After thinking about it for an hour or so, you come up with the following illustration:

Catch-Me Lures, Inc. is a very stable business with expected FCFs of \$1,000 per year, which is likely to be constant for the foreseeable future. The firm currently has no debt outstanding and has an equity beta of 1.0. Given a market risk premium of 5% and a risk-free rate of interest of 8%, you estimate the unlevered cost of equity for the firm to be 8%.

Single-stage valuation; $g=0$

- If Catch-Me Lures' expected FCF is a level perpetuity equal to \$1,000 per year, what is your estimate of the enterprise value of the firm?
- To illustrate the effect of debt financing on the value of Catch-Me Lures, you assume that the firm borrows \$5,000 at a rate of 4% and that the firm pays taxes at a rate of 20%. Compute the amount of taxes the firm will "save" by virtue of the tax deductibility of its interest expense each year.
- It is pretty obvious that borrowing a portion of the firm's capital reduces its tax bill. But how much should you value the interest tax savings due to the use of debt financing? After thinking about it a bit, you decide to implement the basic notion of valuation and discount the future cash flows using a discount rate that reflects the risk of the cash flows. The cash flow estimation was easy; see Exercise 9-4(b). But what discount rate should you use?
- What is the present value of the expected future tax savings from borrowing? What then is the value of the levered firm (i.e., the APV valuation)?

Problem 2

1. $EV_u = \frac{C}{r} = \frac{1,000}{8\%} = 12,500$
2. *Interest tax shield* $= 5,000 \times 4\% \times 20\% = 40$
3. The interest tax savings are computed by multiplying the contractually fixed interest expense (loan amount multiplied by the interest rate) by the firm's tax rate. The primary risk to the firm comes in the form of the risk that the firm will no longer be able to meet its interest payment in a timely manner. Since the lender too faces this risk when determining the borrowing rate of 4%, this seems like a logical approximation to the appropriate discount rate.

Problem 2

$$\text{Enterprise Value (APV Approach)} = \left(\begin{array}{c} \text{Value of the} \\ \text{Unlevered} \\ \text{Free Cash Flows} \end{array} \right) + \left(\begin{array}{c} \text{Value of the} \\ \text{Interest} \\ \text{Tax Savings} \end{array} \right) \quad (9.4)$$

$$4. \quad EV = \frac{1,000}{8\%} + \frac{40}{4\%} = 12,500 + 1000 = 13,500$$

Problem 3

9-5 TRADITIONAL WACC VALUATION The owner of Big Boy Flea Market (BBFM), Lewis Redding, passed away on December 30, 2014. His 100% ownership interest in BBFM became part of his estate, on which his heirs must pay estate taxes. The IRS has hired a valuation expert, who has submitted a report stating that the business was worth approximately \$20 million at the time of Redding's death. Redding's heirs believe the IRS valuation is too high and have hired you to perform a separate valuation analysis in hopes of supporting their position.

- a. Estimate the firm FCFs for 2015 to 2019, where the firm's tax rate is 25%, capital expenditures are assumed to equal depreciation expense, and there are no changes in net working capital over the period.
- b. Value BBFM using the traditional WACC model and the following information:
 1. BBFM's cost of equity is estimated to be 20% and the cost of debt is 10%.
 2. BBFM's management targets its long-term debt ratio at 10% of enterprise value.
 3. After 2019, the long-term growth rate in the firm's free cash flow will be 5% per year.

Problem 3



Given						
Corporate tax rate	25%					
Capex (2010-2014) equals depreciation expense						
Cost of debt financing	10%					
Required return on equity	20%					
Targed debt to value ratio	10%					
Terminal period growth rate in FCFs	5%					
Income Statement	Base Year		Forecast			
	2014	2015	2016	2017	2018	2019
Sales	\$ 3,417,500.00	\$ 3,700,625.00	\$ 3,983,750.00	\$ 4,266,875.00	\$ 4,550,000.00	\$ 4,833,125.00
Depreciation		48,190.60	50,118.22	52,122.95	54,207.87	56,376.19
COGS		1,402,798.69	1,496,881.13	1,590,963.56	1,685,046.00	1,779,128.44
Other Operating Expenses		1,078,447.38	1,109,874.25	1,141,301.13	1,172,728.00	1,204,154.88
EBIT		\$ 1,171,188.34	\$ 1,326,876.40	\$ 1,482,487.36	\$ 1,638,018.13	\$ 1,793,465.50
Interest		5,110.00	5,110.00	5,110.00	5,110.00	5,860.00
Earnings before Taxes		\$ 1,166,078.34	\$ 1,321,766.40	\$ 1,477,377.36	\$ 1,632,908.13	\$ 1,787,605.50
Income Taxes		(291,519.58)	(330,441.60)	(369,344.34)	(408,227.03)	(446,901.38)
Net Income		\$ 874,558.75	\$ 991,324.80	\$ 1,108,033.02	\$ 1,224,681.10	\$ 1,340,704.13
Preferred Stock Dividends		-	-	-	-	-
Net Income to Common Stock		874,558.75	991,324.80	1,108,033.02	1,224,681.10	1,340,704.13
Common Stock Dividends		874,558.75	991,324.80	1,108,033.02	1,224,681.10	1,340,704.13
Income to Retained Earnings		-	-	-	-	-
Balance Sheet	Base Year		Forecast			
	2014	2015	2016	2017	2018	2019
Current Assets	\$ 170,630.00	\$ 185,031.25	\$ 199,187.50	\$ 213,343.75	\$ 227,500.00	\$ 241,656.25
Gross Fixed Assets	1,204,765.00	1,252,955.60	1,303,073.82	1,355,196.78	1,409,404.65	1,540,780.83
Accumulated Depreciation	(734,750.00)	(782,940.60)	(833,058.82)	(885,181.78)	(939,389.65)	(995,765.83)
Net Fixed Assets	\$ 470,015.00	\$ 470,015.00	\$ 470,015.00	\$ 470,015.00	\$ 470,015.00	\$ 545,015.00
Total Assets	\$ 640,645.00	\$ 655,046.25	\$ 669,202.50	\$ 683,358.75	\$ 697,515.00	\$ 786,671.25
Current Liabilities	\$ 129,645.00	\$ 144,046.25	\$ 158,202.50	\$ 172,358.75	\$ 186,515.00	\$ 200,671.25
L-T Debt	51,100.00	51,100.00	51,100.00	51,100.00	51,100.00	58,600.00
Preferred Stock	-	-	-	-	-	-
Common Stock	1,000.00	1,000.00	1,000.00	1,000.00	1,000.00	68,500.00
Retained Earnings	458,900.00	458,900.00	458,900.00	458,900.00	458,900.00	458,900.00
Total Common Equity	\$ 459,900.00	\$ 459,900.00	\$ 459,900.00	\$ 459,900.00	\$ 459,900.00	\$ 527,400.00
Total Liabilities & Equity	\$ 640,645.00	\$ 655,046.25	\$ 669,202.50	\$ 683,358.75	\$ 697,515.00	\$ 786,671.25

Problem 3

	Base Year	
	2014	2015
Income Statement		
Sales	\$ 3,417,500.00	\$ 3,700,625.00
Depreciation		48,190.60
COGS		1,402,798.69
Other Operating Expenses		1,078,447.38
EBIT		\$ 1,171,188.34
Interest		5,110.00
Earnings before Taxes		\$ 1,166,078.34
Income Taxes		(291,519.58)
Net Income		\$ 874,558.75
Preferred Stock Dividends		-
Net Income to Common Stock		874,558.75
Common Stock Dividends		874,558.75
Income to Retained Earnings		-
Balance Sheet		
	2014	2015
Current Assets	\$ 170,630.00	\$ 185,031.25
Gross Fixed Assets	1,204,765.00	1,252,955.60
Accumulated Depreciation	(734,750.00)	(782,940.60)
Net Fixed Assets	\$ 470,015.00	\$ 470,015.00
Total Assets	\$ 640,645.00	\$ 655,046.25
Current Liabilities	\$ 129,645.00	\$ 144,046.25
L-T Debt	51,100.00	51,100.00
Preferred Stock	-	-
Common Stock	1,000.00	1,000.00
Retained Earnings	458,900.00	458,900.00
Total Common Equity	\$ 459,900.00	\$ 459,900.00
Total Liabilities & Equity	\$ 640,645.00	\$ 655,046.25

a. Estimate firm free cash flows for 2014-2019

	2015
Net Operating Income	\$ 1,171,188.34
Less: Taxes	(292,797.08)
NOPAT	\$ 878,391.25
Plus: Depreciation	48,190.60
Less: Capex	(48,190.60)
Less: Change in NWC	\$ -
Firm Free Cash Flow (FCF)	\$ 878,391.25

25%

=782,940.60-734,750.00

Problem 3

$$1. \quad FCF_1 = 878,391.25; FCF_2 = 995,157.30; FCF_3 = 1,111,865.52; FCF_4 = 1,228,513.60; FCF_5 = 1,345,099.13$$

$$WACC = k_d(1 - T)w_d + k_pw_p + k_e w_e \quad (4.1)$$

$$2. \quad WACC = 10\% \times (1 - 25\%) \times 10\% + 20\% \times 90\% = 18.75\%$$

$$\bullet \quad PP_0 = \frac{878,391.25}{(1+18.75\%)^1} + \frac{995,157.30}{(1+18.75\%)^2} + \frac{1,111,865.52}{(1+18.75\%)^3} + \frac{1,228,513.60}{(1+18.75\%)^4} + \frac{1,345,099.13}{(1+18.75\%)^5} = 3,296,797.37$$

Problem 3

$$\frac{\text{Terminal Value}_{2020}}{\text{Terminal Value}_{2020}} = \frac{\text{FCF}_{2020} \left(1 + \frac{\text{Terminal Growth Rate}(g)}{\text{Terminal Growth Rate}(g)} \right)}{\left(\frac{\text{Cost of Capital}}{(k_{WACC})} - \frac{\text{Terminal Growth Rate}(g)}{\text{Terminal Growth Rate}(g)} \right)} \quad (9.2)$$

- $TV_5 = \frac{1,345,099.13 \times (1+5\%)}{(18.65\% - 5\%)} = 10,271,666.05$
- $TV_0 = \frac{10,271,666.05}{(1+18.75\%)^5} = 4,349,835.17$
- $EV = PP_0 + TV_0 = 3,296,797.37 + 4,349,835.17 = 7,646,632.53$
- Conclusion: We estimate that the enterprise value of BBFM is \$7,646,632.53 which is substantially **less than the \$20 million** estimate that came from the IRS. Note, however, that we assume that the firm has no cash or other non-operating assets.

- *Thank you for your time and attention!*